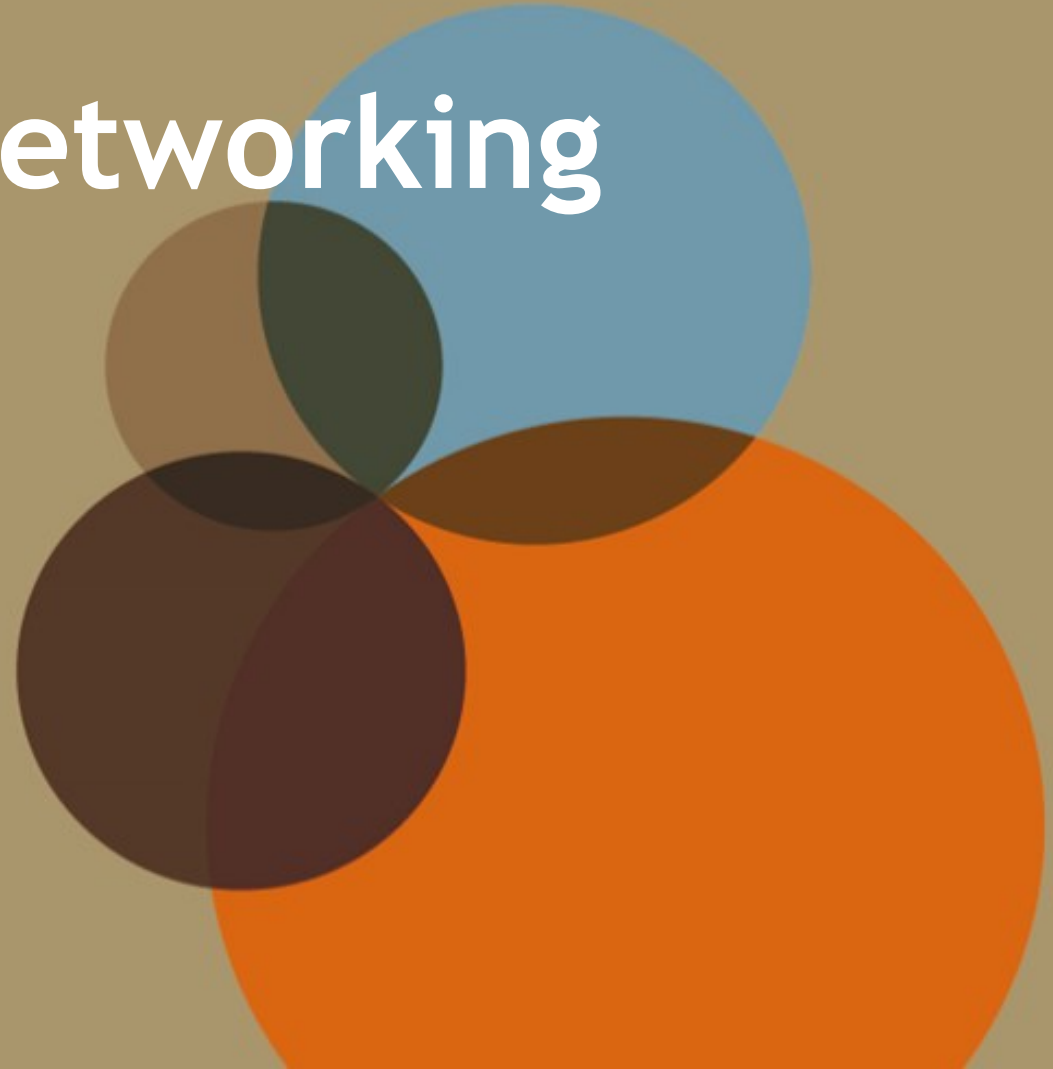


Computer Networks

Chapter 4

Advanced Internetworking

Prof. Marino Miculan



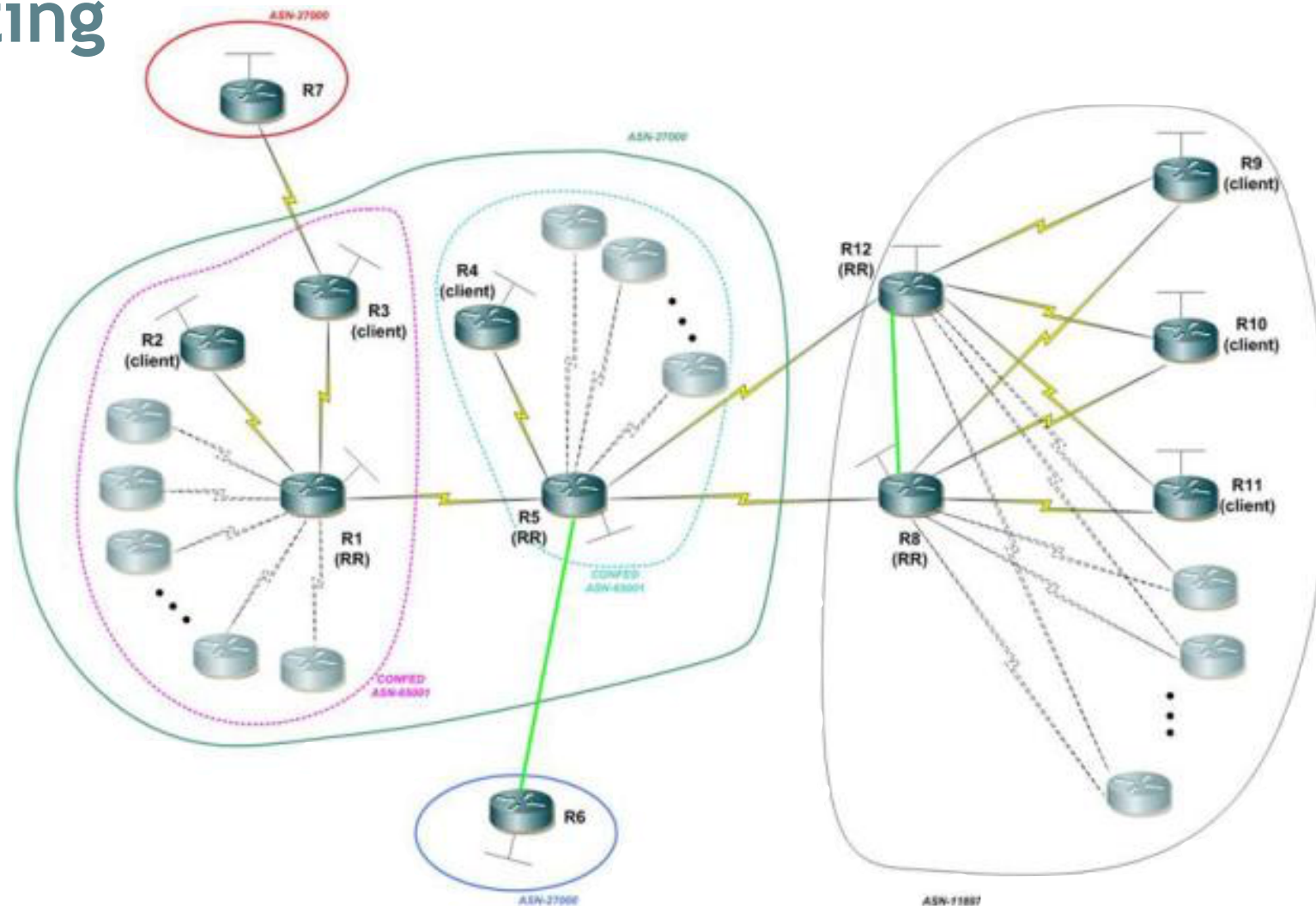
Problems

- How do we build a routing system that can handle hundreds of thousands of networks and billions of end nodes?
- How to handle address space exhaustion of IPV4?
- How to enhance the functionalities of Internet?

Interdomain Routing

- Internet is organized as **autonomous systems (AS)** each of which is under the control of a single administrative entity
- Autonomous System (AS)
 - corresponds to an administrative domain
 - examples: University, company, backbone network
- A corporation's internal network might be a single AS, as may the network of a single Internet service provider

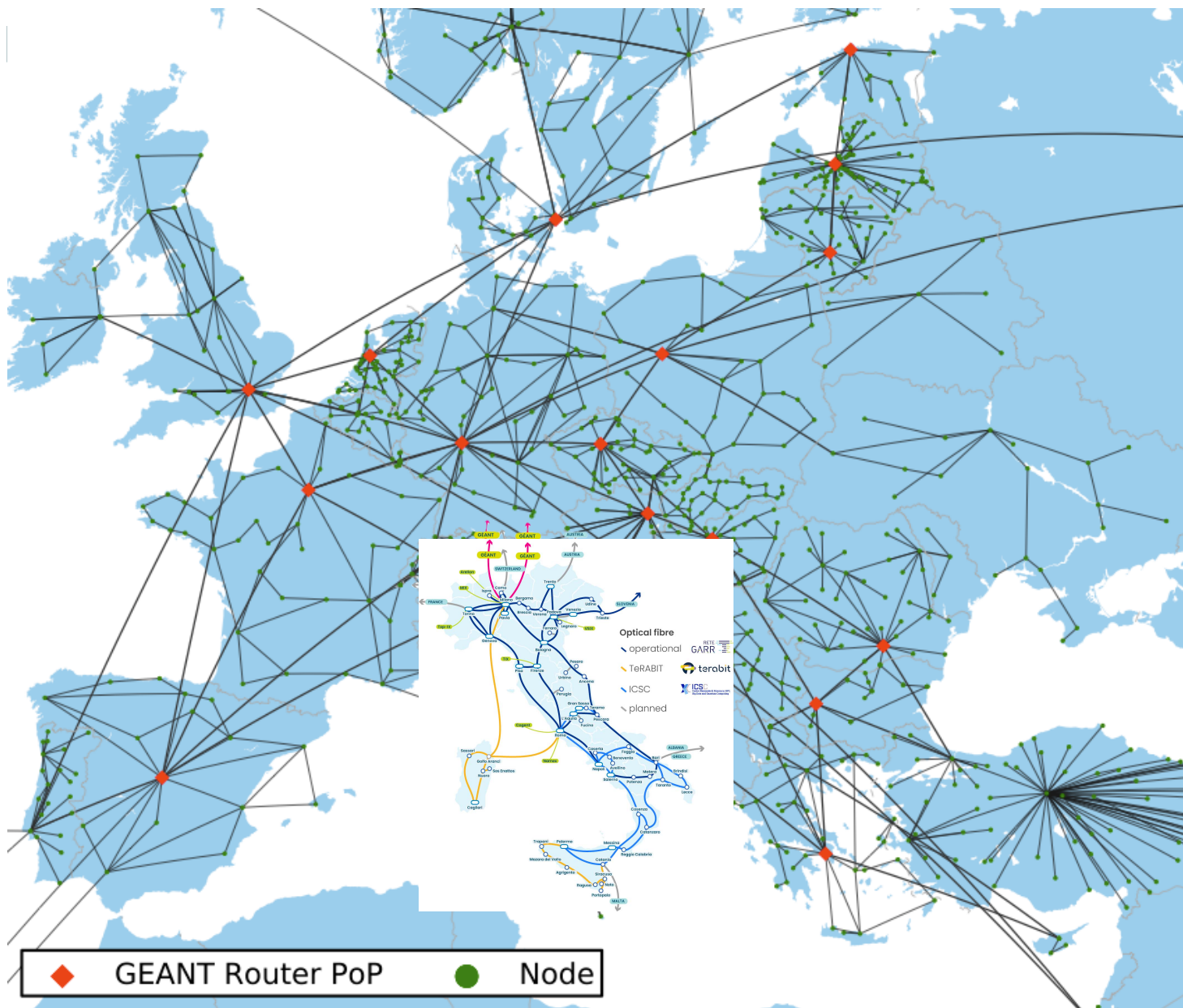
Interdomain Routing



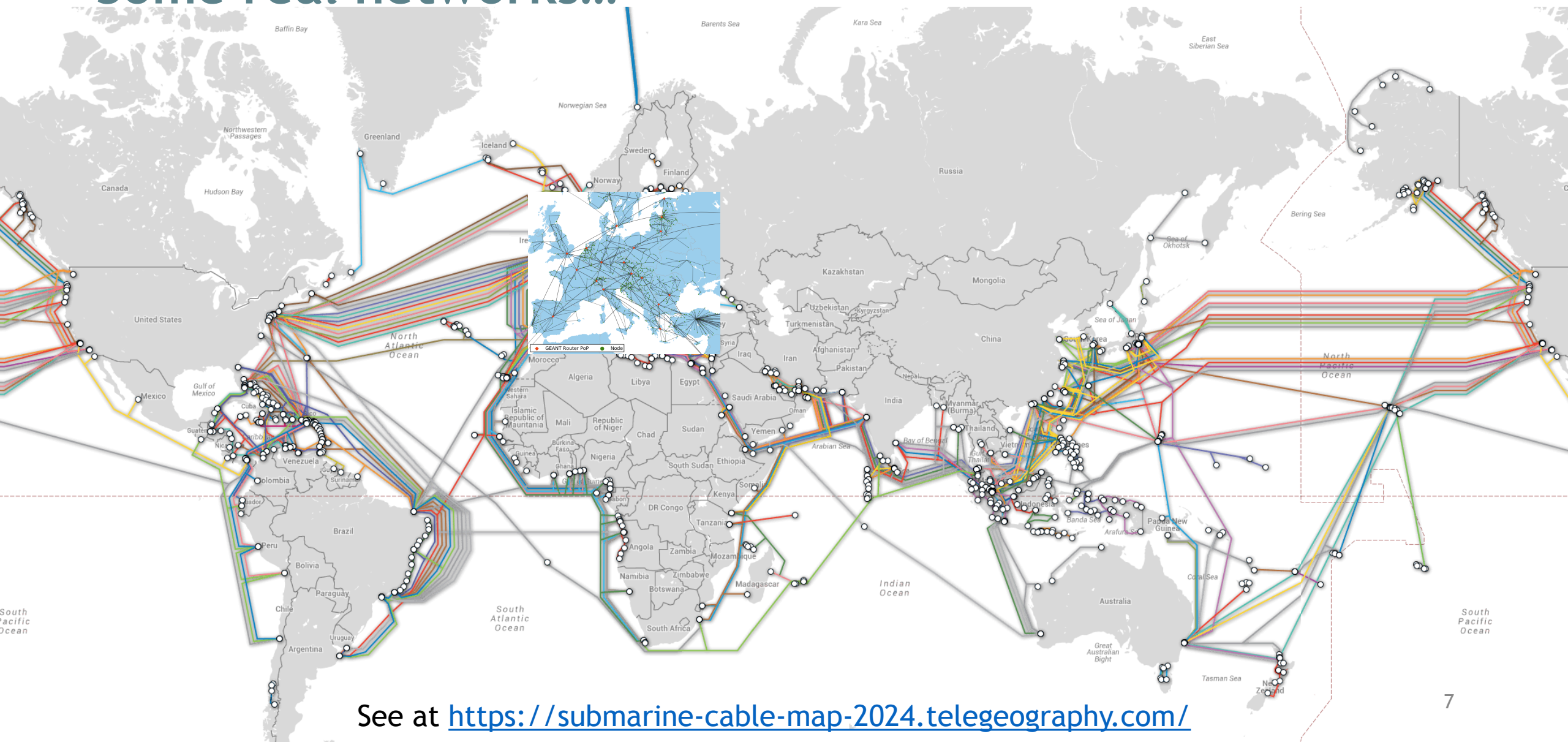
A network with some autonomous system

□

Some real



Some real networks...



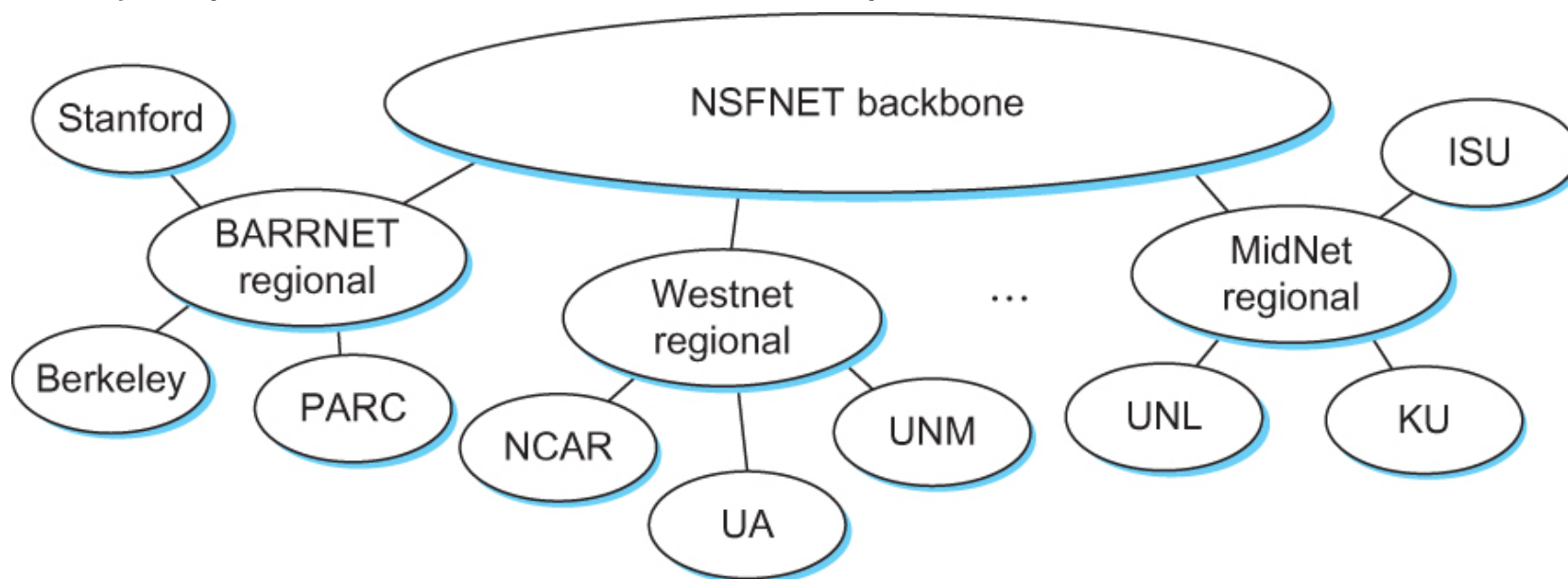
See at <https://submarine-cable-map-2024.telegeography.com/>

Route Propagation

- Idea: Provide an additional way to hierarchically aggregate routing information in a large internet.
 - Improves scalability
- Divide the routing problem in two parts:
 - Routing **within** a single autonomous system
 - Routing **between** autonomous systems
- Another name for autonomous systems in the Internet is *routing domains*
- Two-level route propagation hierarchy
 - Inter-domain routing protocol (Internet-wide standard)
 - Intra-domain routing protocol (each AS selects its own)

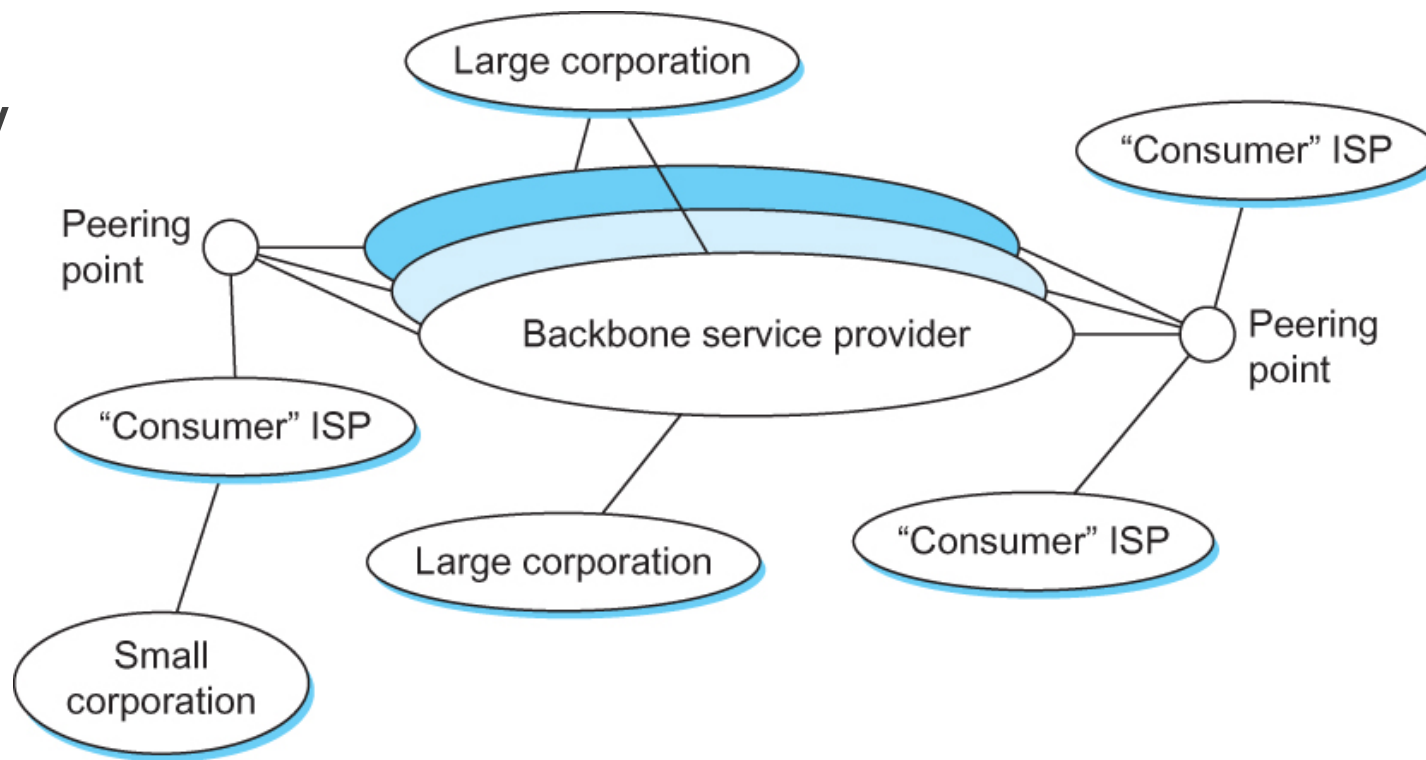
EGP and BGP

- Inter-domain Routing Protocols
 - Exterior Gateway Protocol (EGP)
 - Forced a tree-like topology onto the Internet
 - Did not allow for the topology to become general
 - Tree like structure: there is a single backbone and autonomous systems are connected only as parents and children and not as peers - used to be like this in 1990:



Border Gateway Protocol (BGP)

- Assumes that the Internet is an arbitrarily interconnected set of ASs.
- Today's Internet consists of an interconnection of multiple backbone networks (they are usually called service provider networks, and they are operated by private companies rather than the government)
- Sites are connected to each other in arbitrary ways



BGP

- Some large corporations connect directly to one or more of the backbone, while others connect to smaller, non-backbone service providers.
- Many service providers exist mainly to provide service to “consumers” (individuals with PCs in their homes), and these providers must connect to the backbone providers
- Often many providers arrange to interconnect with each other at a single “peering point”

BGP-4: Border Gateway Protocol

- Assumes the Internet is an arbitrarily interconnected set of AS's.
- Define **local traffic** as traffic that originates at or terminates on nodes within an AS, and **transit traffic** as traffic that passes through an AS.
- We can classify AS's into three types:
 - **Stub AS**: an AS that has only a single connection to one other AS; such an AS will only carry local traffic (small corporation in the figure of the previous page).
 - **Multihomed AS**: an AS that has connections to more than one other AS, but refuses to carry transit traffic (large corporation at the top in the figure of the previous page).
 - **Transit AS**: an AS that has connections to more than one other AS, and is designed to carry both transit and local traffic (backbone providers).

BGP

- The goal of Inter-domain routing is to find any path to the intended destination that is **loop free**
 - We are concerned with reachability than optimality
 - Finding path anywhere close to optimal is considered to be a great achievement
- Why?

BGP

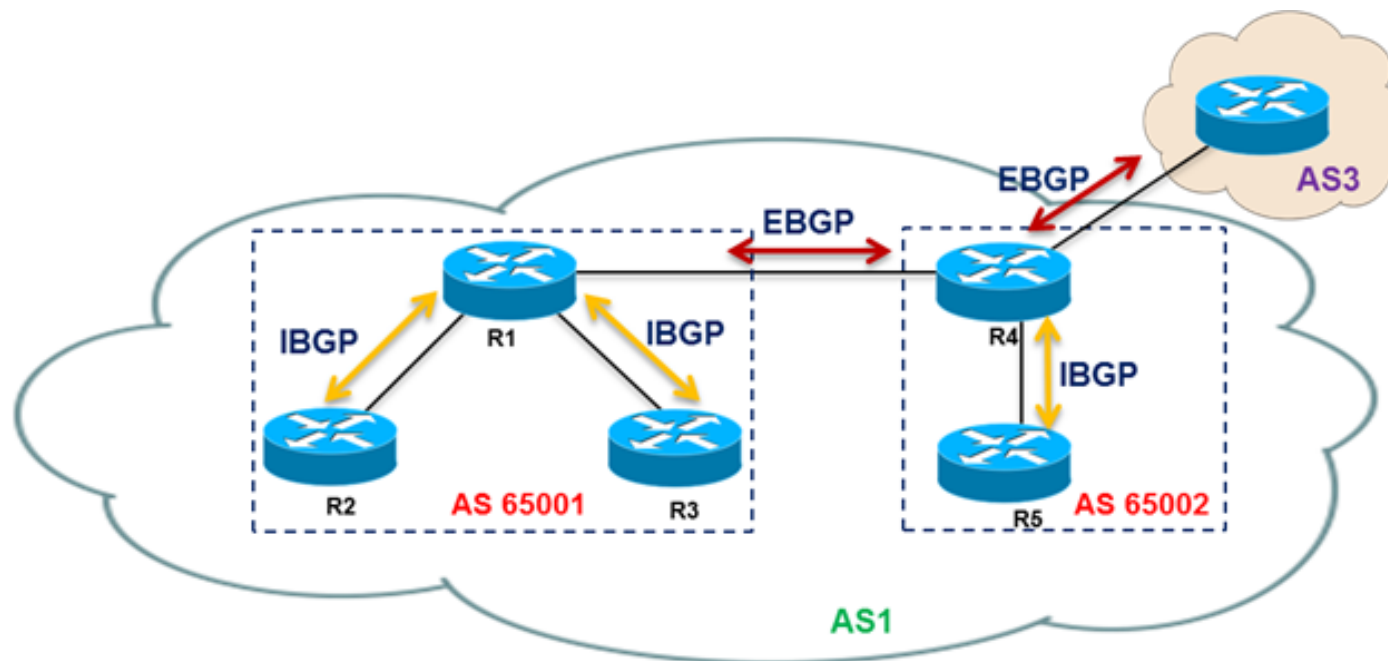
- **Scalability:** An Internet backbone router must be able to forward any packet destined anywhere in the Internet
 - Having a routing table that will provide a match for any valid IP address
- **Autonomous** nature of the domains
 - It is impossible to calculate meaningful path costs for a path that crosses multiple ASs
 - A cost of 1000 across one provider might imply a great path but it might mean an unacceptable bad one from another provider
- Issues of **trust**
 - Provider A might be unwilling to believe certain advertisements from provider B

BGP

- Each AS has one (or more) **BGP speaker** that advertises to its peers:
 - local networks
 - other reachable networks (transit AS only)
 - gives path information
- In addition to the BGP speakers, the AS has one or more **border gateways** which need not be the same as the speakers
- The border gateways are the routers through which packets enter and leave the AS

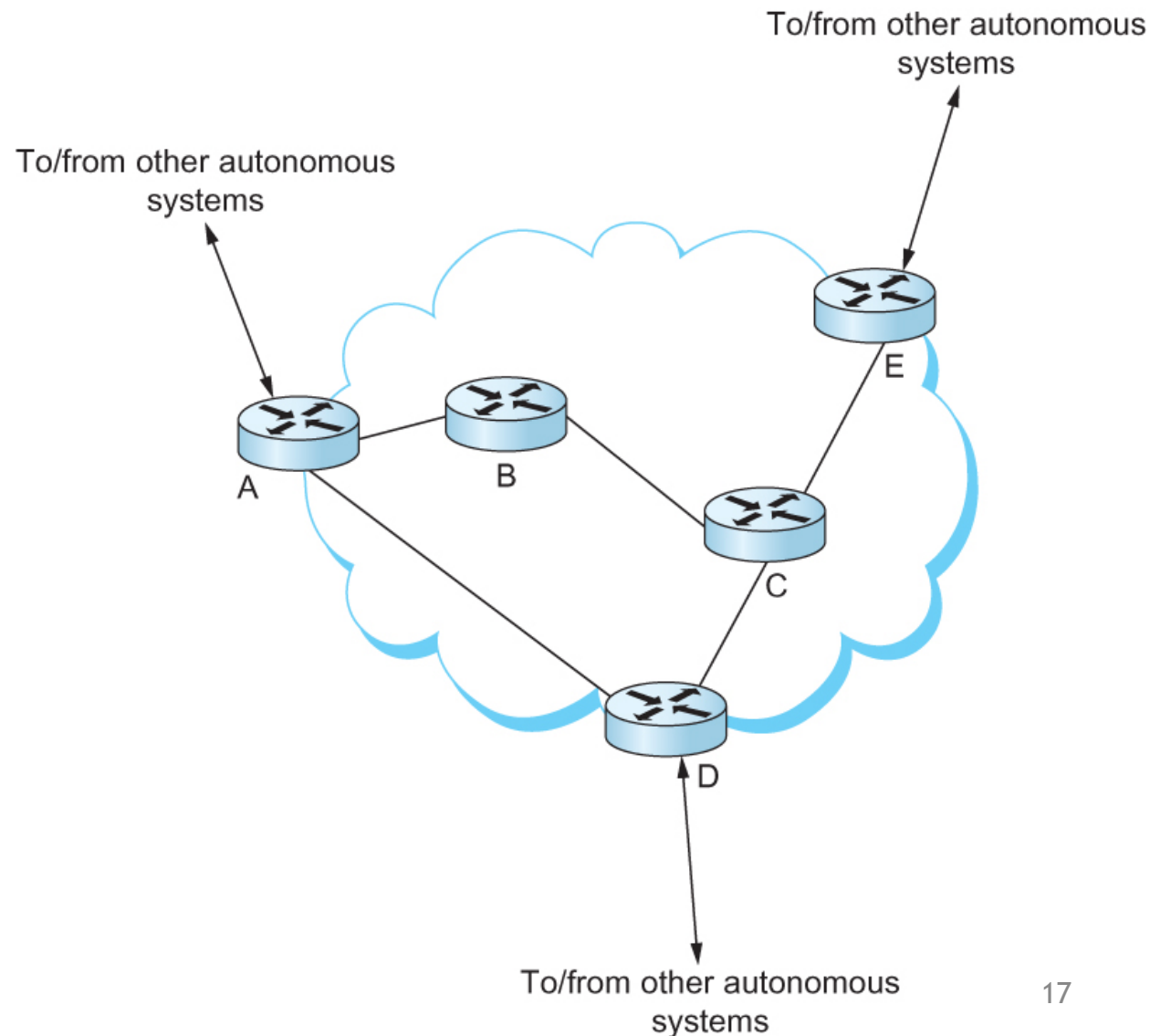
eBGP and iBGP

- Two versions of protocol:
 - **External BGP (eBGP)**: used between routers belonging to different ASs
 - **Internal BGP (iBGP)**: used between routers inside the same ASs



Integrating Interdomain and Intradomain Routing

- All routers run iBGP and an intradomain routing protocol (e.g. RIP, OSPF, ...).
- Border routers (A, D, E) also run eBGP to other ASs
- There is a BGP routing table valid for the whole AS, whose information is then propagated to the internal routers via iBGP



Integrating Interdomain and Intradomain Routing

Prefix	BGP Next Hop
18.0/16	E
12.5.5/24	A
128.34/16	D
128.69./16	A

BGP Table for the AS

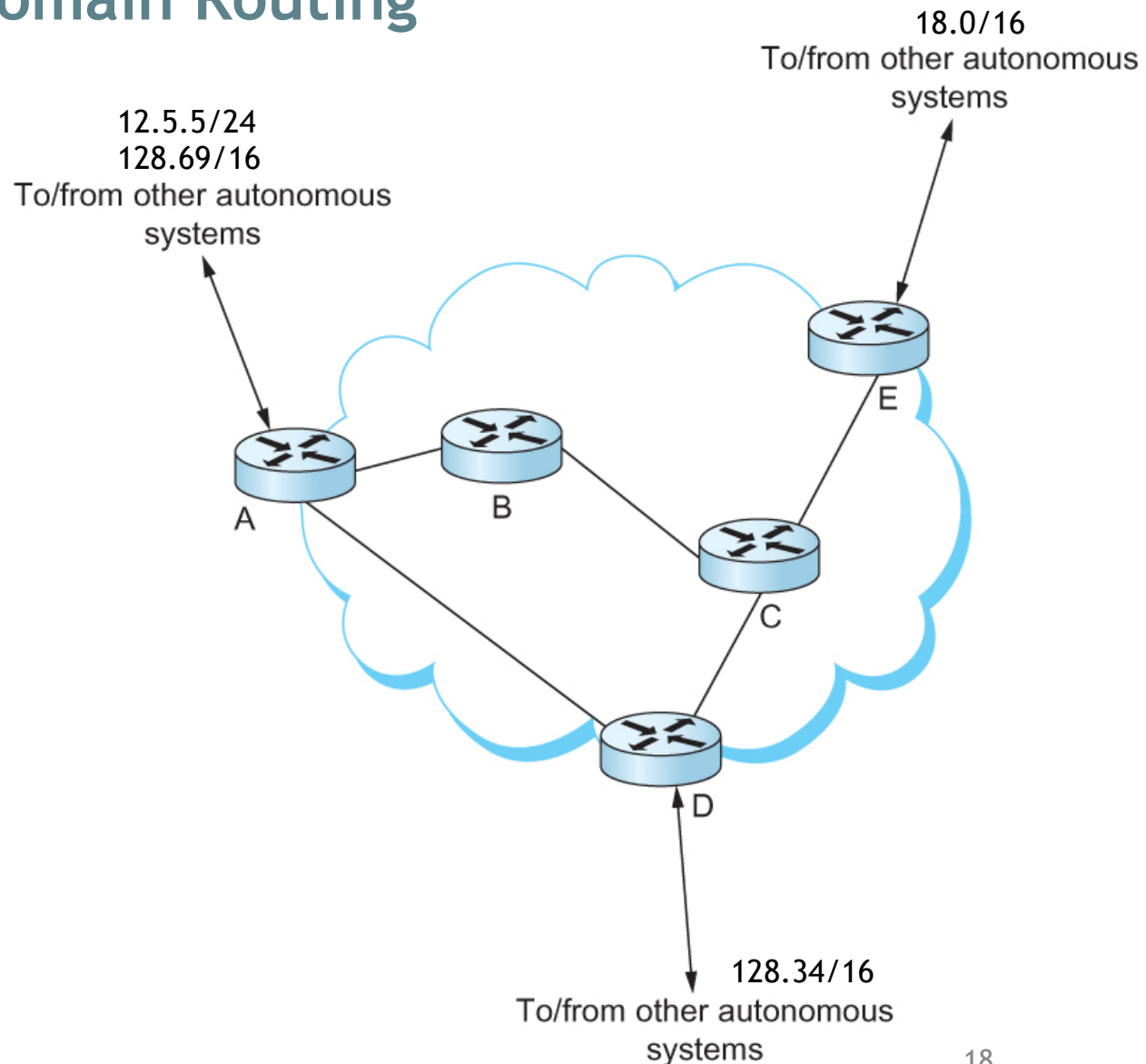
Router	IGP Path
A	A
C	C
D	C
E	C

IGP Table for Router B

Prefix	IGP Path
18.0/16	C
12.5.5/24	A
128.34/16	C
128.69./16	A

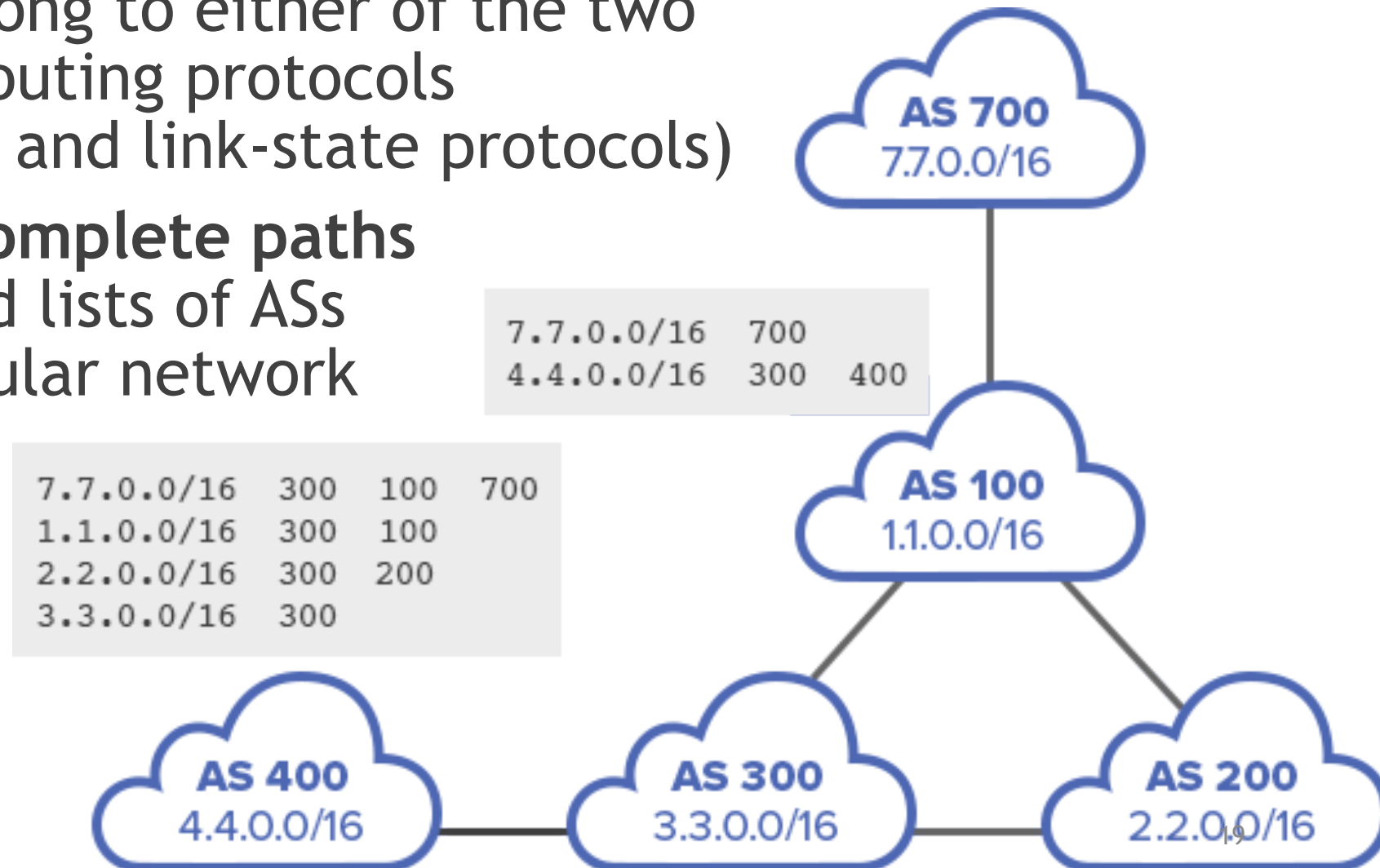
Combined Table for Router B

BGP routing table, IGP routing table,
and combined table at router B



BGP

- BGP does not belong to either of the two main classes of routing protocols (distance vectors and link-state protocols)
- BGP advertises **complete paths** as an enumerated lists of ASs to reach a particular network



BGP loop prevention and policy routing

- A BGP router ignores a path if
 - its AS appears in it (this avoids loops between ASs)
 - Or if it violates some policy of the AS

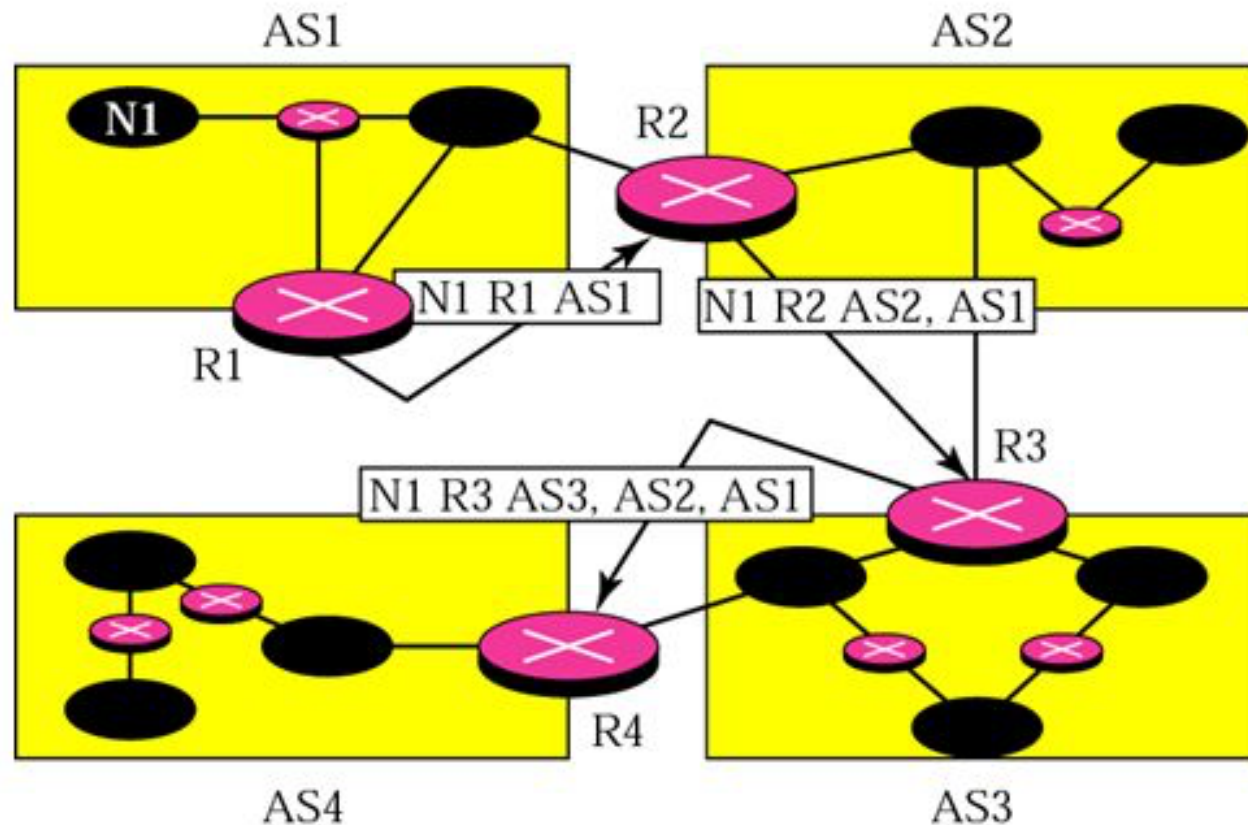
BGP loop prevention and policy routing

Example of Network Reachability

Network	Next router	Path
N1	R1	AS14,AS23,AS67
N2	R5	AS22,AS67,AS5,AS89
N3	R6	AS67,AS89,AS9,AS34
N4	R12	AS62,AS2,AS9

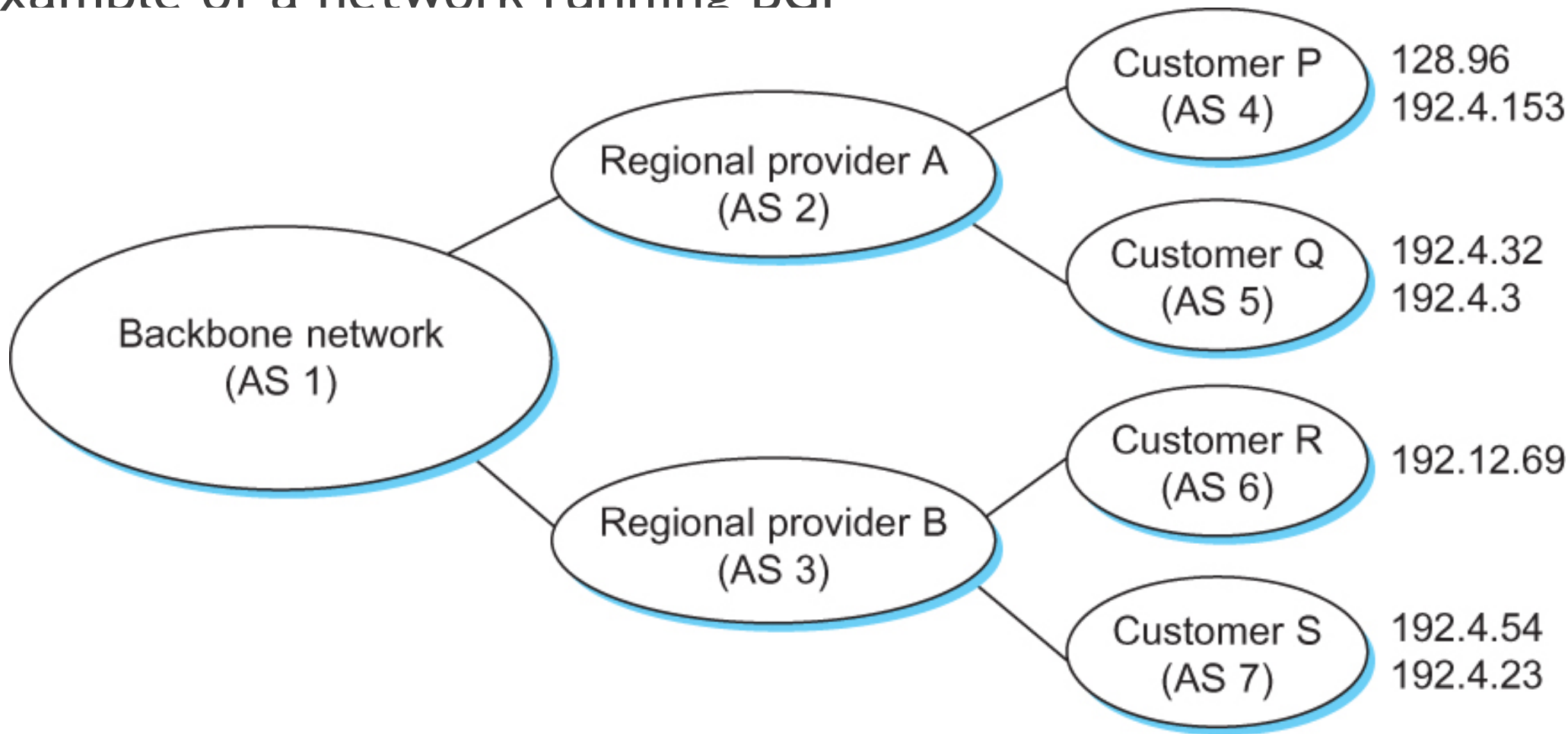
- Loop Prevention in BGP:
 - Checks the Path before updating its database. (If its AS is in the path ignore the message)
- Policy Routing:
 - If a path consist of an AS against the policy of the current AS, message discarded.

Example of Message advertisements



BGP Example

- Example of a network running BGP



BGP Example

- Speaker for AS 2 advertises reachability to P and Q
 - Network 128.96, 192.4.153, 192.4.32, and 192.4.3, can be reached directly from AS 2.
- Speaker for backbone network then advertises
 - Networks 128.96, 192.4.153, 192.4.32, and 192.4.3 can be reached along the path <AS 1, AS 2>.
- Speaker can also cancel previously advertised paths

BGP Issues

- AS numbers carried in BGP need to be unique
 - For example, AS 2 can only recognize itself in the AS path in the example if no other AS identifies itself in the same way
- AS numbers (ASN) are 32-bit numbers assigned by a central authority (IANA and regional internet registries) (RFC 4893)
- About 60.000 ASs, at the moment
 - these are the nodes of the network running eBGP

Next Generation IP (IPv6)

- IPv4 addresses are running out!

Nikolas Pediaditis [npediaditi at ripe.net](mailto:npediaditi@ripe.net)

Mon Nov 25 15:39:12 CET 2019

Dear colleagues,

Today, at 15:35 UTC+1 on 25 November 2019, we made our final /22 IPv4 allocation from the last remaining addresses in our available pool. We have now run out of IPv4 addresses.

Our announcement will not come as a surprise for network operators – IPv4 run-out has long been anticipated and planned for by the RIPE community. In fact, it is due to the community's responsible stewardship of these resources that we have been able to provide many thousands of new networks in our service region with /22 allocations after we reached our last /8 in 2012.

Next Generation IP (IPv6)

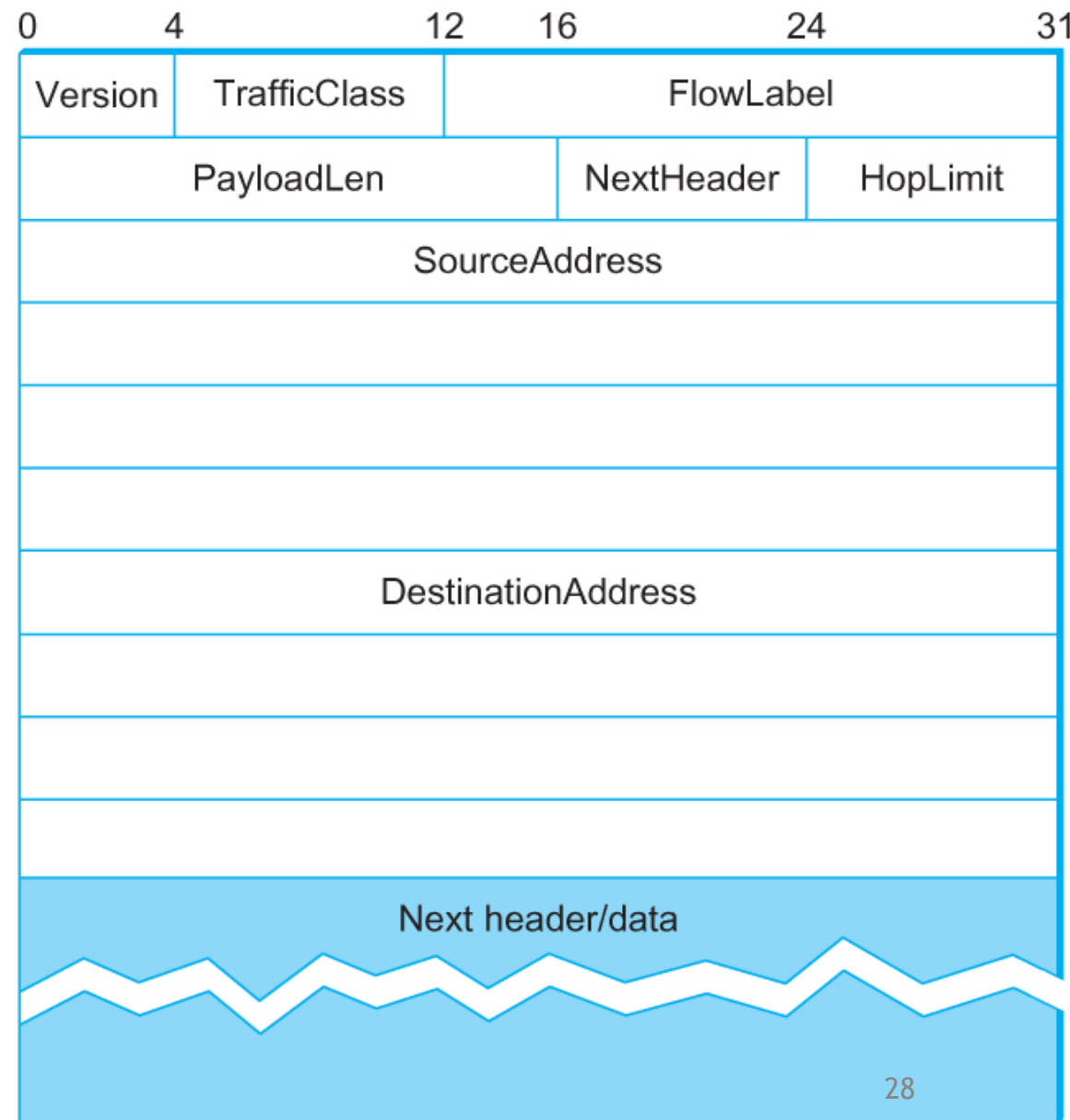
- 128-bit addresses
 - Address space in the order of $3 \cdot 10^{38}$.
 - As a comparison: Earth surface = $5,1 \cdot 10^{14} \text{ m}^2 = 5,1 \cdot 10^{20} \text{ mm}^2$, so $0,58 \cdot 10^{18}$ addresses for each mm^2
- Multicast
- Real-time service
- Authentication and security
- Auto-configuration
- End-to-end fragmentation
- Enhanced routing functionality, including support for mobile hosts
- Extensible

IPv6 Addresses

- Classless addressing/routing (similar to CIDR)
- Notation: x:x:x:x:x:x:x:x (x = 16-bit hex number)
 - Example: 47CD:0000:0000:0000:0000:0000:A456:0124
 - contiguous 0s are compressed: 47CD::A456:124
 - IPv6 compatible IPv4 address: ::128.42.1.87
 - This does not mean that a IPv4 host can receive IPv6 packets!
 - Localhost: 0000:0000:0000:0000:0000:0000:0000:0001, i.e. ::1
- Address assignment
 - provider-based (as for IPv4)
 - geographic

IPv6 Header

- 40-byte “base” header
- 64Kbyte payload (with extensions)
- Extension headers (fixed order, mostly fixed length)
 - fragmentation
 - source routing
 - authentication and security
 - and many other



Priority

- *Traffic Class* defines the priority of a datagram with respect to others of the same source
- Useful in case some datagrams have to be dropped due to congestions
- Possible values:
 - 0=Generic traffic
 - 1=Background traffic (e.g. NNTP)
 - 2=Traffic without waiting (e.g. SMTP)
 - 3 e 5=reserved
 - 4=Traffic with waiting (e.g. FTP, HTTP)
 - 6=Interactive traffic (e.g. TELNET, SSH)
 - 7=Control traffic (OSPF, RIP, SNMP,...)
 - 8-15=Traffic not depended on congestion (it should not be dropped), ordered by redundancy (e.g. audio/video)

Flow Label

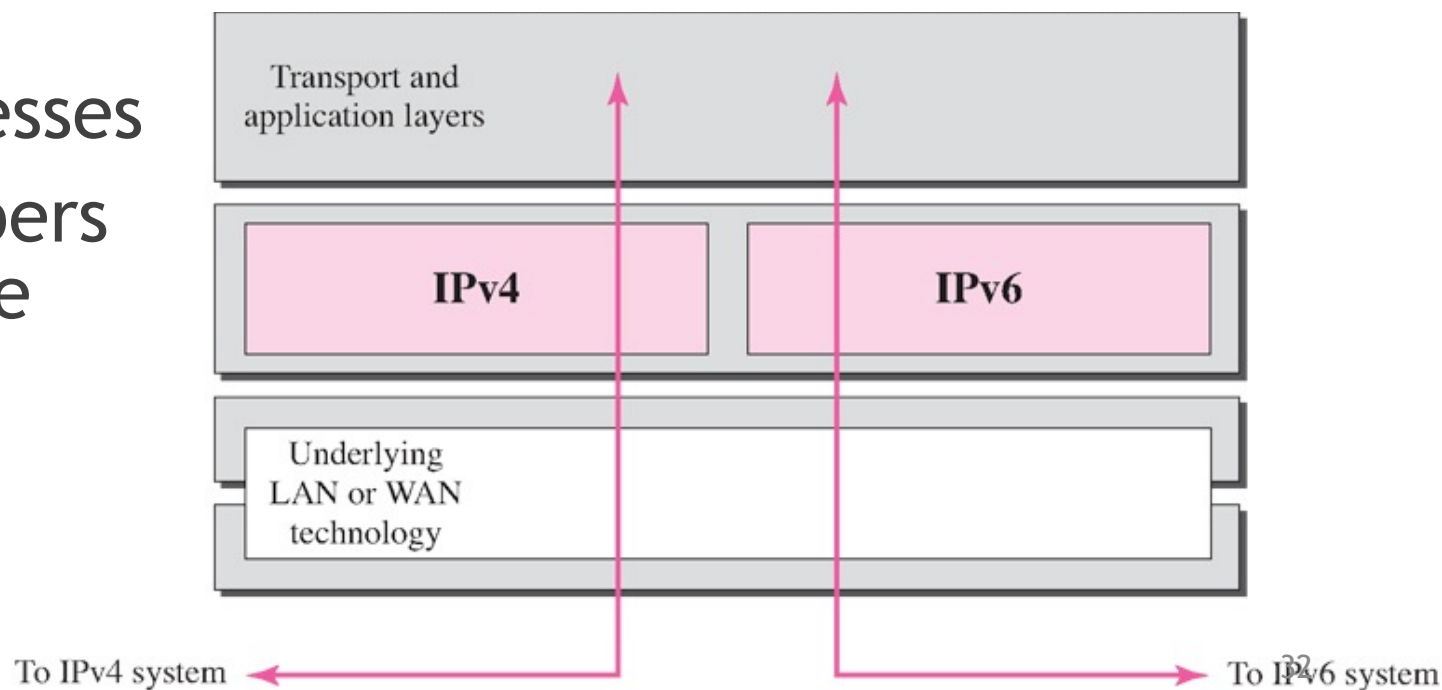
- 20 bit casual number, assigned by source
- Datagrams belonging to the same flow are labelled with the same value
- This label can be used by routers for implementing a coherent service to all datagrams of the same flow. E.g.:
 - same routing
 - Quality of Service: usage of reserved resources (previously reserved by means of other protocols, e.g. RSVP)
 - useful for soft real-time

IPv4 to IPv6 transition

- IPv4 and IPv6 are not compatible
- Eventually IPv4 should be replaced by IPv6
- Very hard. Too many hosts to update
 - Cannot be done at once
 - Cannot be forced “by law”
- 3 techniques to support transition
 - Dual protocol stack
 - Tunneling
 - Header translation

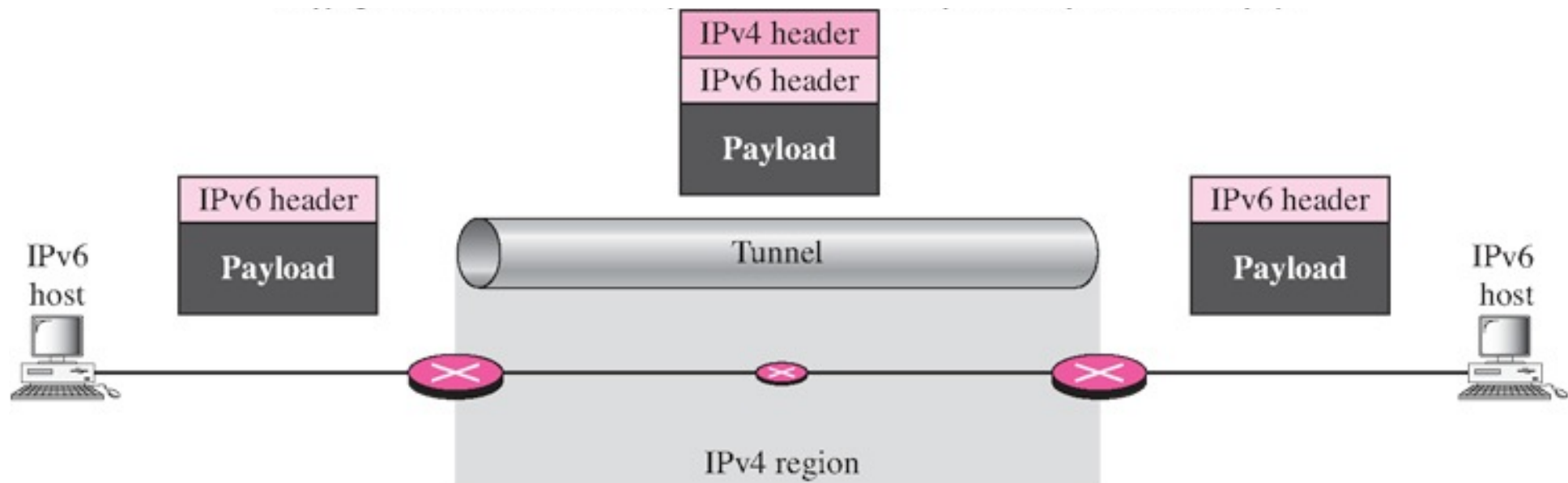
Dual protocol stack

- A host (or router) can implement both protocols
 - Used in most modern operating systems
- Two protocols at level 3: the host belongs to two separated internetworks
- Applications have to deal explicitly with both addresses
 - more work for developers and deployers (e.g. see Apache `httpd.conf`)



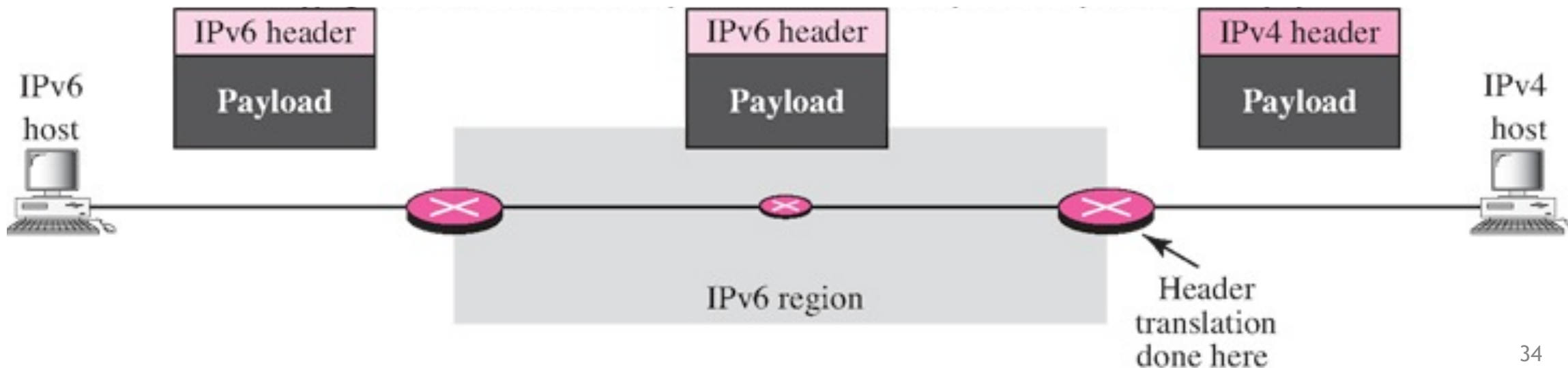
Tunneling

- Create “tunnels” or “bridges” from one IPv6 region to another, incapsulating IPv6 datagrams in IPv4 datagrams
- Managed transparently by routers
- Hosts can communicate in IPv6
- Benefits of IPv6 are lost on the IPv4 region



Header translation

- Allows to transmit an IPv6 datagram to an IPv4-only host
- Destination router replaces IPv6 header with IPv4 “equivalent”, and vice versa
- Similar to NAT - but translation is between IPv4-IPv6 addresses and headers

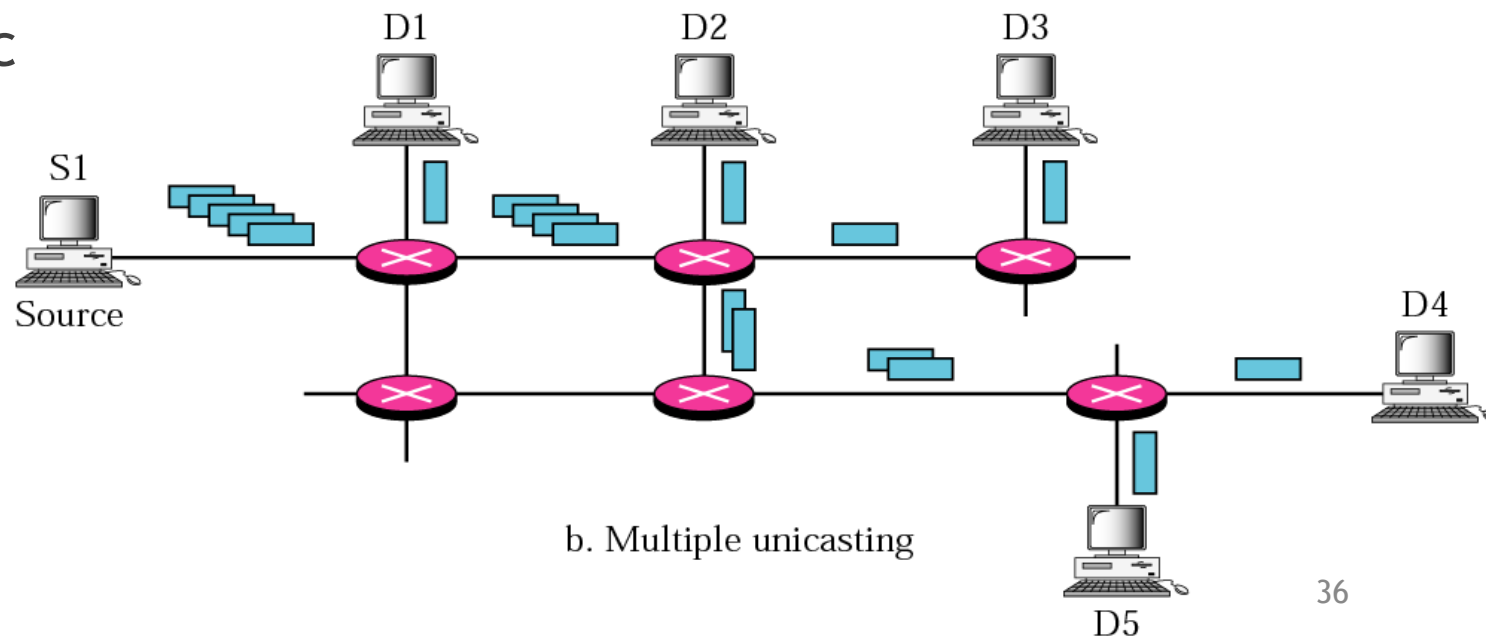


Internet Multicast

- Multicast = the possibility of sending a message to a group of receiving hosts, without knowing or specifying them
- One-to-many: Source specific multicast (SSM)
 - A receiving host specifies a multicast group where a specific host is sending. Examples:
 - Radio station broadcast
 - Transmitting news, stock-price
 - Software updates to multiple hosts
- Many-to-many: Any source multicast (ASM)
 - Multimedia teleconferencing
 - Online multi-player games
 - Distributed simulations

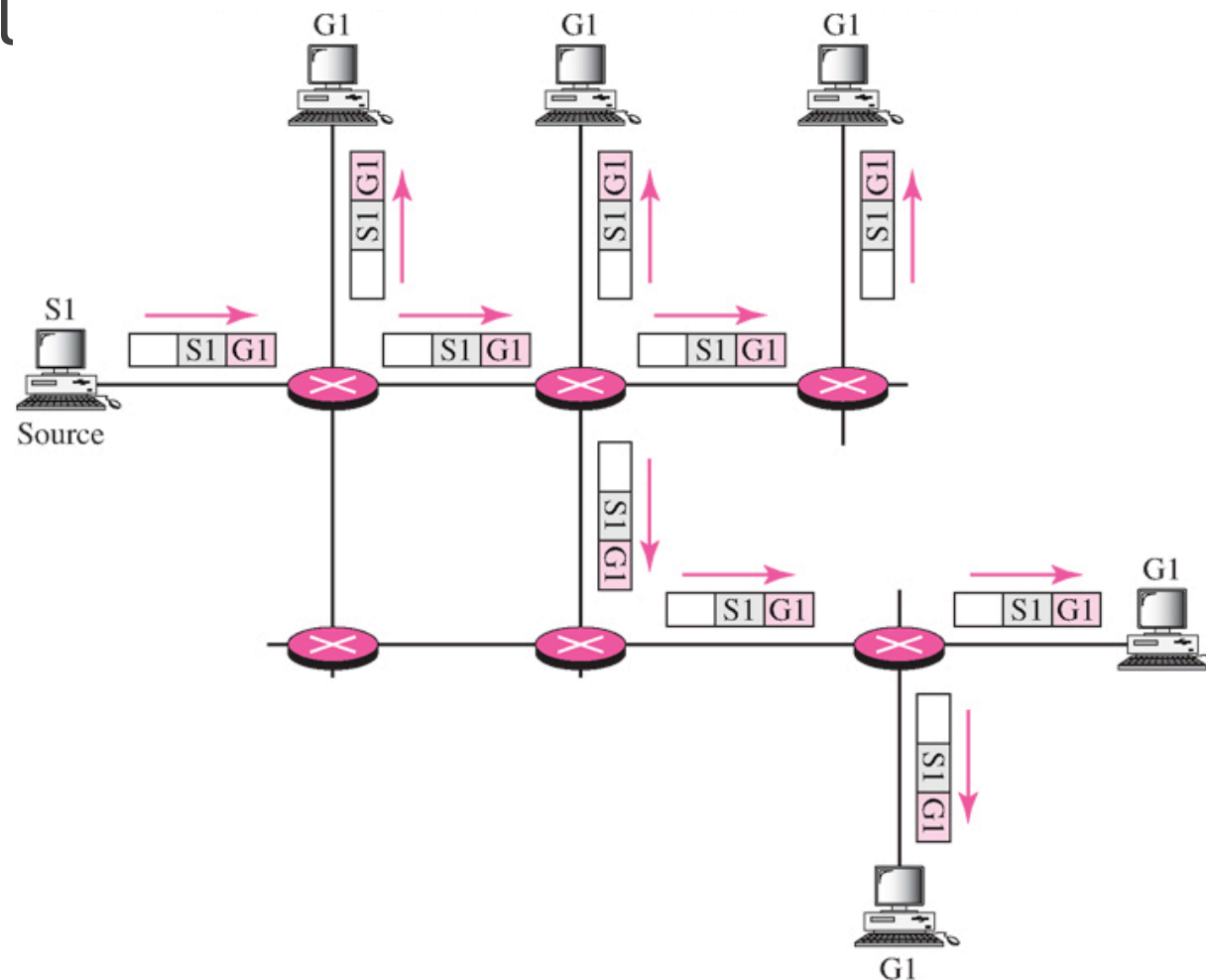
Multicast without multicast?

- Without support for multicast, a source needs to send a separate packet with the identical data to each member of the group
 - This redundancy consumes more bandwidth
 - Redundant traffic is not evenly distributed, concentrated near the sending host
 - Source needs to keep track of the IP address of each member in the group
 - Group may be dynamic
- Differences of delays between different destinations are increased



Multicast in IP

- Using IP multicast to send the identical packet to each member of the group
 - A host sends a *single* copy of the packet, addressed to the group's multicast address
 - Each host members of that group receives a copy of the packet
 - The source host does not need to know the individual unicast IP address of each member
 - Packets are duplicated by routers along the way, when needed



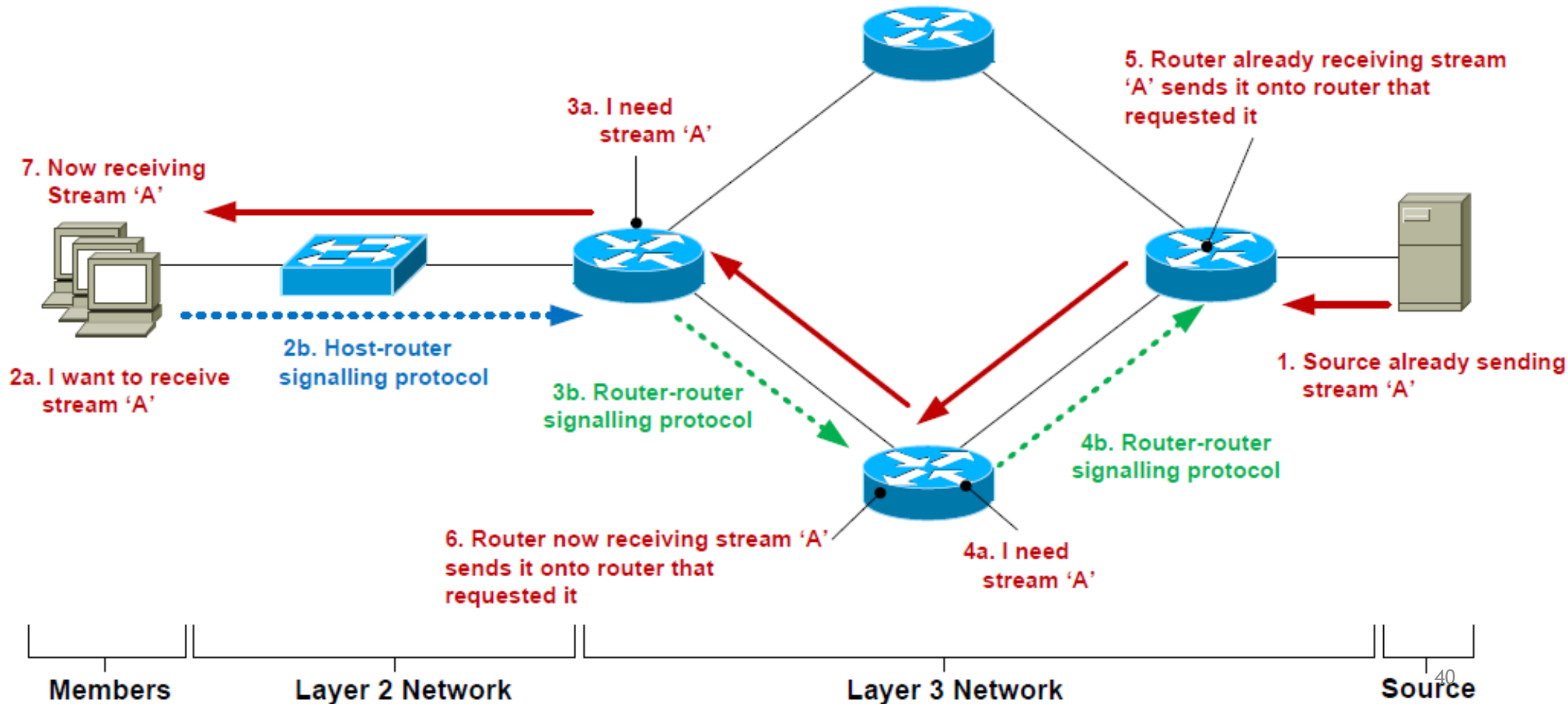
Multicast in IP

- IP provides an IP-level many-to-many multicast model based on **multicast groups**
 - Each group has its own IP multicast address in the D class (224.0.0.0 - 239.255.255.255)
 - Group addresses are assigned in many ways:
 - 224.0.0.0/24: Use is defined by IANA, but restricted to local network
 - 224.0.1.0/24: Global groups, statically assigned by IANA; e.g. 224.0.1.1 is NTP
 - Dynamically leased for the time of a *session*, using SAP/SDP protocol
 - ...

Multicast group management

- A host can join and leave groups
- A host can be in multiple groups
- A host signals its desire to join or leave a multicast group by communicating with its *local router* using a special protocol
 - In IPv4: Internet Group Management Protocol (**IGMP**)
 - In IPv6: Multicast Listener Discovery (**MLD**)
- The router has the responsibility for making multicast behave correctly with regard to the host

Multicast Service Model Overview



Multicast service

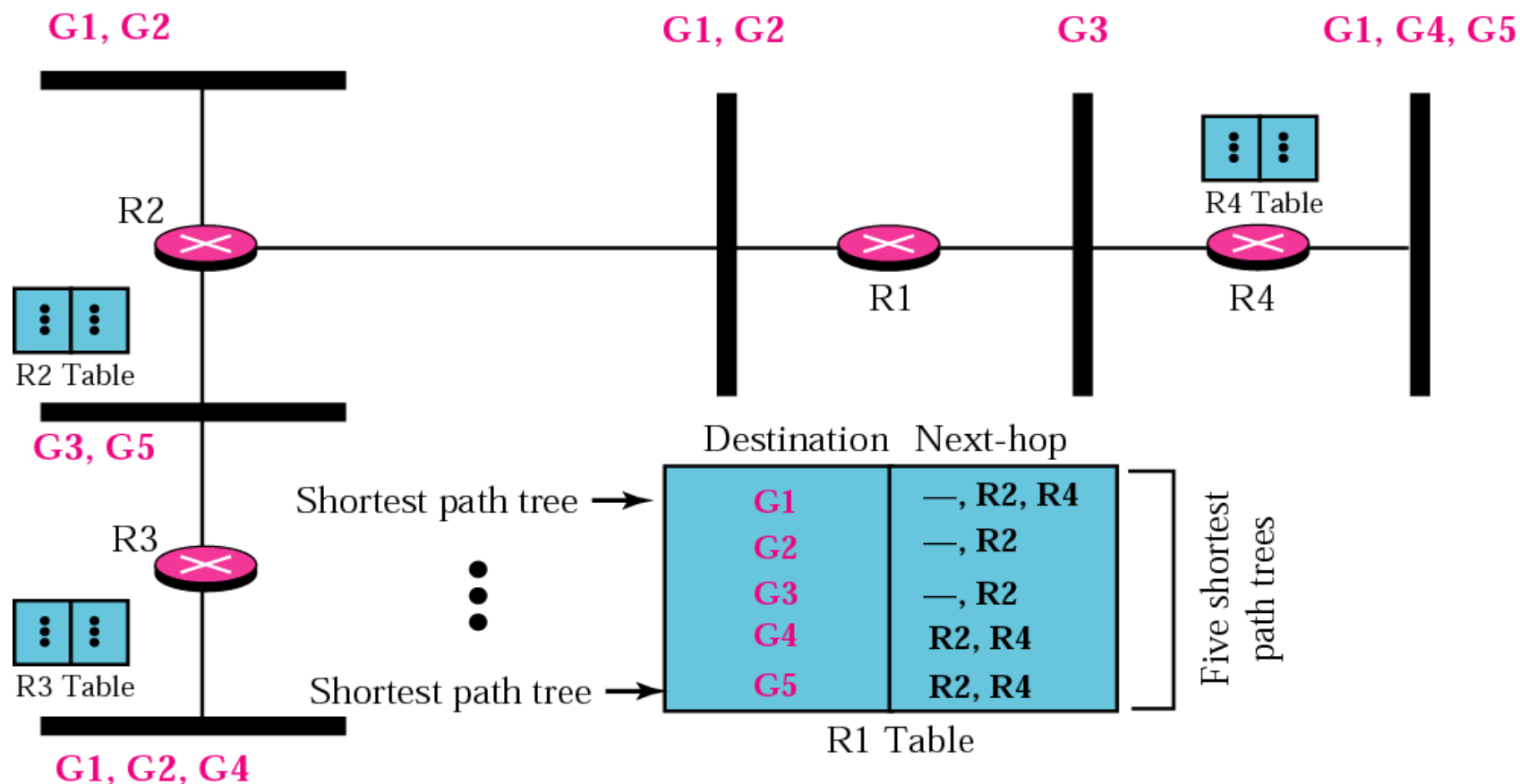
- Once a host has joined a group, it receives all messages sent to that group, and can send any message to that group as well
- Scope of messages can be defined by choosing a suitable TTL
- Security issues (secrecy, integrity, authentication, access control...) are not handled at this level (can be handled at session/application level)

Multicast Routing

- A router's unicast forwarding tables indicate for any IP address, which link to use to forward the unicast packet
- To support multicast, a router must additionally have **multicast forwarding tables** that indicate, based on multicast destination address, which links to use to forward the multicast packet
- Multicast forwarding tables collectively specify a set of trees, one for each group: **Multicast distribution trees**
- **Multicast routing** is the process by which multicast distribution trees are determined
- Many routing protocols have been proposed: DVMRP, RPB, PIM...
- Two families: **source-based** and **group-shared** trees

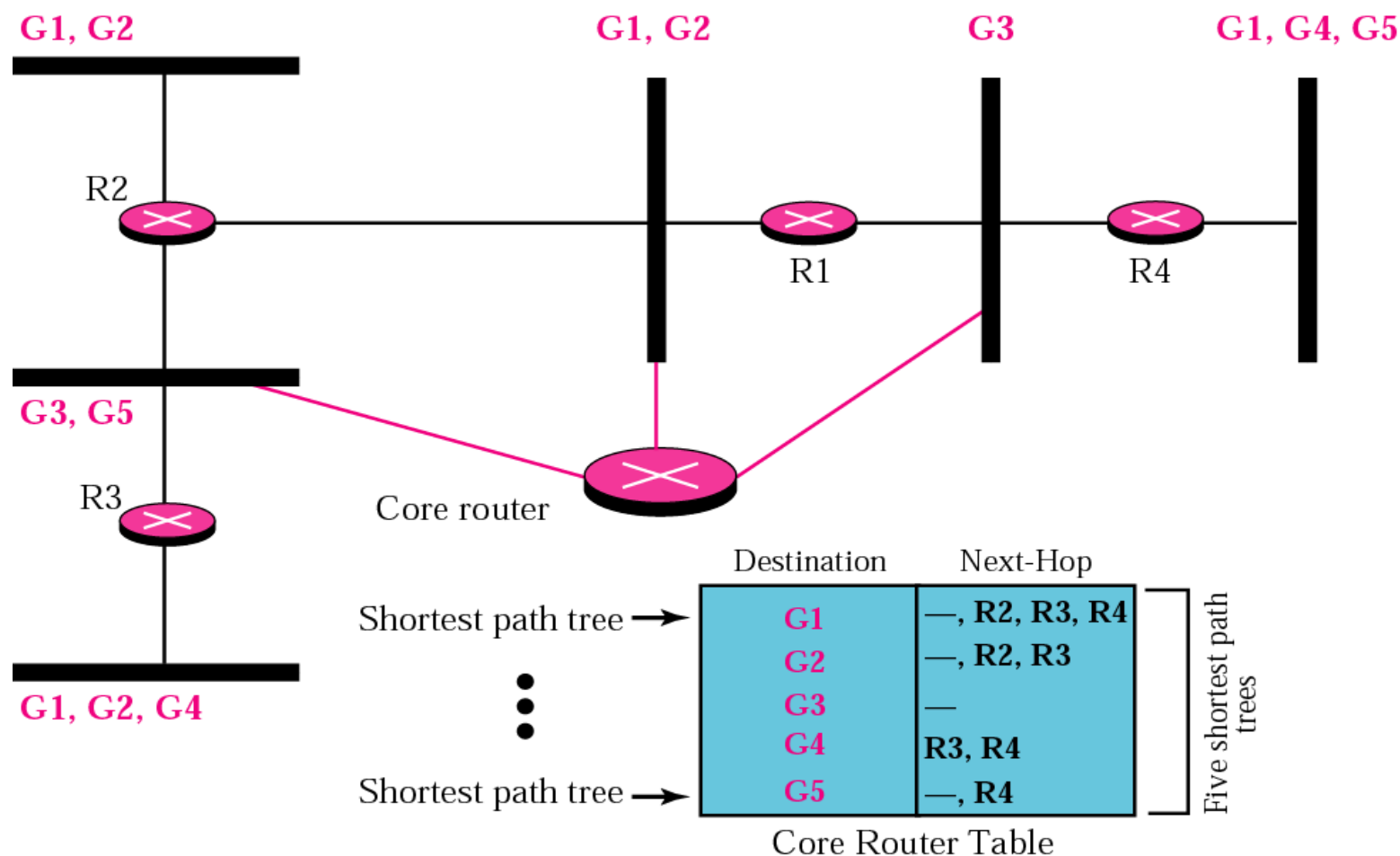
Source-based multicast routing

- In the **source-based tree** approach, each router needs to have one shortest path tree for each group.



Group-shared tree multicast routing

- In the group-shared tree approach, only a router (called **core** or **rendezvous** router) has a shortest path tree for each group, and is involved in multicasting.

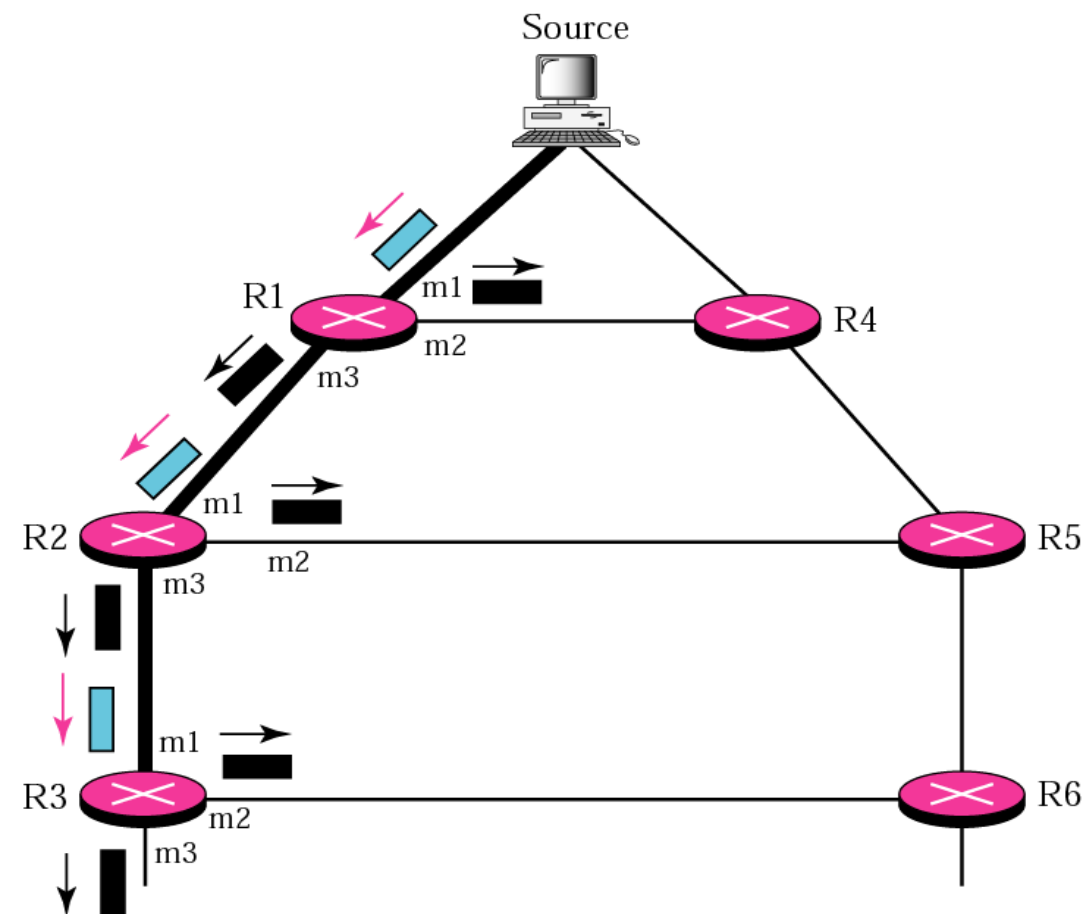


Distance-Vector Multicast Routing Protocol (DVMRP)

- First multicast routing protocol implemented (RFC 1075, Nov. 1988)
- Extends distance vector routing protocols for unicast routing
- “Flood and prune” strategy: flood the network with multicast traffic, and in the meanwhile prune the branches not interested in the traffic

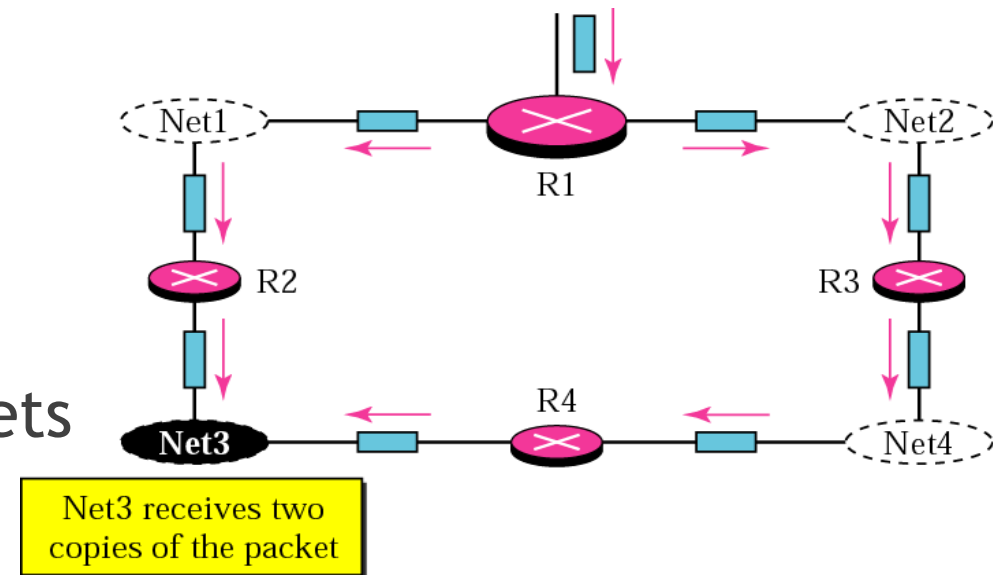
Distance-Vector Multicast Routing Protocol (DVMRP)

- Flooding part: **Reverse Path Flooding**
 - From the unicast routing table, each router already knows the next hop which the shortest path to a source S goes through.
 - When receives a multicast packet, a router looks at its source address S , and then forwards it on all outgoing links (except the one on which the packet arrived), **if and only if** packet arrived from the next hop towards S (i.e., in “reverse path” direction).
 - In this way, there are no loops in the flooding process



Distance-Vector Multicast Routing Protocol (DVMRP)

- Still, networks with more than one router can get duplicate broadcast packets
- Eliminate duplicate packets by fixing a “parent” (relative to S) router for each LAN and letting only parents forward packets
- Parent routers are those with the shortest path to S (which they know via distance vector) - in the case above, R2 is parent for Net3
- Break ties by choosing router with smallest IP address
 - (Compare with the Spanning tree protocol in switched LANs)
- This strategy, called **Reverse Path Broadcast**, guarantees that each LAN receives exactly one copy of each packet (less loss of packets), along a **source-based shortest path tree**



Distance-Vector Multicast Routing Protocol (DVMRP)

- **Pruning part:** Prune networks that have no hosts interested in group G. Done in two steps.
- Step 1: Determine if LAN is a leaf of the distribution tree with no members in G
 - A LAN is a leaf if its parent is the only router on the LAN
 - Hosts in the LAN willing to participate to G must notify the router using IGMP (periodically - they are removed after a timeout)
 - Thus, the router knows if LAN has no members of G

Distance-Vector Multicast Routing Protocol (DVMRP)

- Step 2: Propagate “no members of G here” information
 - augment each entry <Destination, Cost> in distance vectors sent to neighbours with the list of groups for which this network is interested in receiving multicast packets
 - Thus, each neighbour knows whether they have to consider this router in the reverse path flooding/broadcast, by looking at the groups it is interested into
 - Including always the list of interested groups in distance vectors may be heavy and useless (e.g. in the case a LAN is interested in several groups, but no host is transmitting on these groups)
 - Instead, the router adds to distance vectors the list of groups which it is NOT interested into, and only when multicast address becomes active (someone is transmitting)

Distance-Vector Multicast Routing Protocol (DVMRP)

- So, when a multicast transmission starts on group G:
 - at first it floods all the network, via RPF/RPB, building a distribution tree which covers basically all the network (because no one is saying that it is not interested in G, yet)
 - and then the tree is pruned: the routers not interested in G start notifying back to stop transmitting to them
 - If a host of a pruned net joins the group G later on, it notifies its local router using IGMP, and the router will start accepting the traffic from his neighbours (*grafting*)
- Overall, DVMRP works well on small scale, less on large scale, due to this continuous flooding-and-pruning

Multicast routing: PIM

- Protocol Independent Multicast
 - independent from the construction of the network topology (to be solved with a different protocol)
- Has two different modes
- PIM-DM (dense mode) (RFC 3973, 2005)
 - Used when many hosts are origin of data with respect to the routers
 - e.g. inside a LAN, or an autonomous system
- PIM-SM (sparse mode) (RFC 2362, 1998)
 - Used when multicast involves few nodes in the net (with respect to the number of routers)
 - E.g. few hosts across the Internet

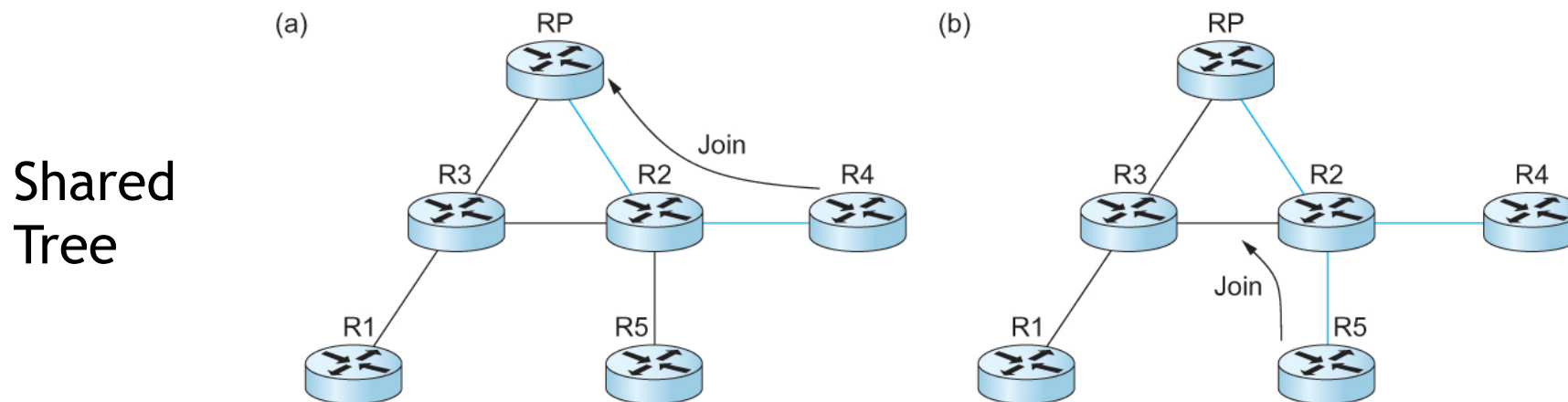
PIM Dense Mode

- Similar to DVMRP, but independent from the underlying unicast routing protocol (differently from DVMRP, which is tied to distance vector protocols)
- **Source-based trees:** each node (router) keeps its copy of multicast distribution trees
- Uses flood-and-prune strategy: RPF to distribute traffic, and pruning/grafting to handle trees.
- Works well when the group is not too large, and not too many routers have to be traversed (e.g. inside a LAN or AS)

PIM Sparse Mode

- Can scale to large groups, sparse on the Internet
 - (e.g. it is used for IPTV)
- For each group, a **group-shared tree** is built and in each AS a “rendez-vous” router is elected
- Routers (in consequence of an IGMP request from some local host) can join the shared trees by sending a (unicast) “join” request to their rendez-vous point
- As it traverses the routers, the “join” message to group G creates a multicast distribution tree, rooted at the rendez-vous point RP for G

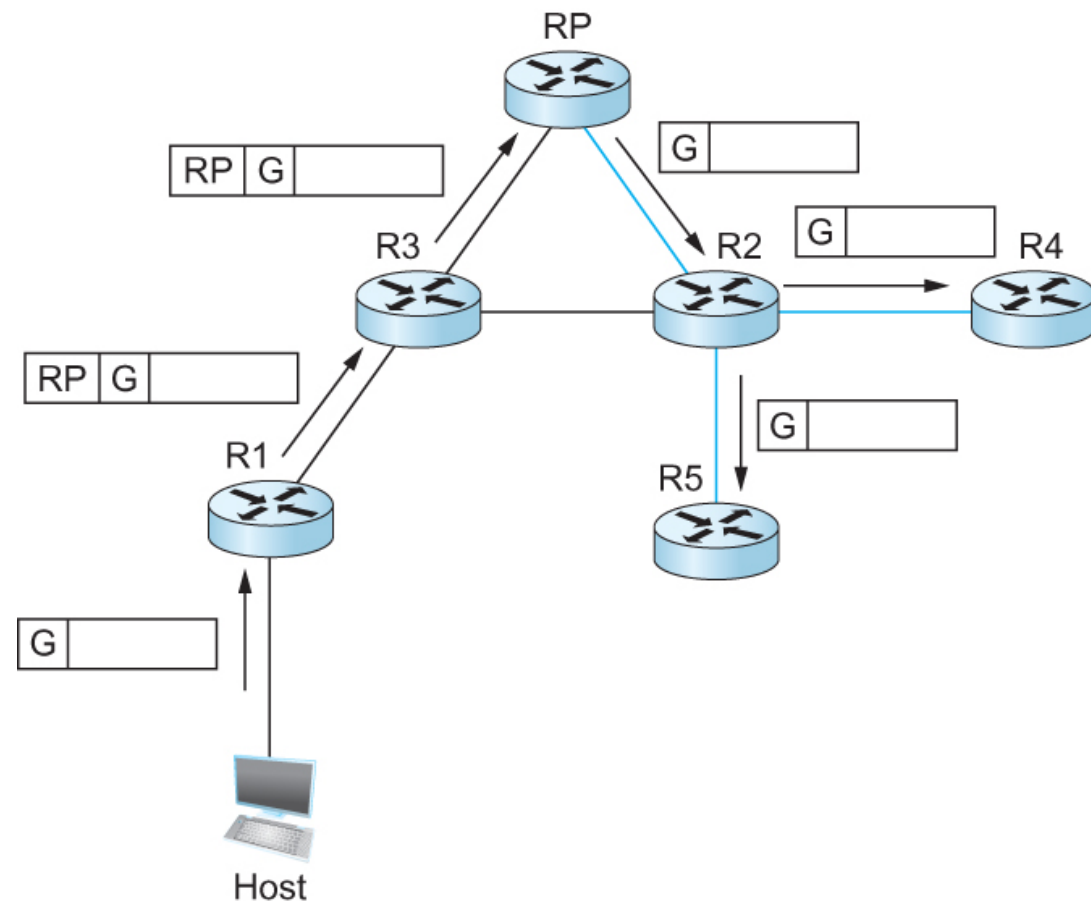
PIM Sparse Mode



- Each router analyses the message and adds to its table the rule to forward downwards the traffic from group G along the interface the “join” message came from
- If the router was not participating to the tree already (a), then it forwards the “join” request towards RP, and marks the corresponding interface as the only one where the traffic can come from; otherwise (b) does nothing.

PIM Sparse Mode

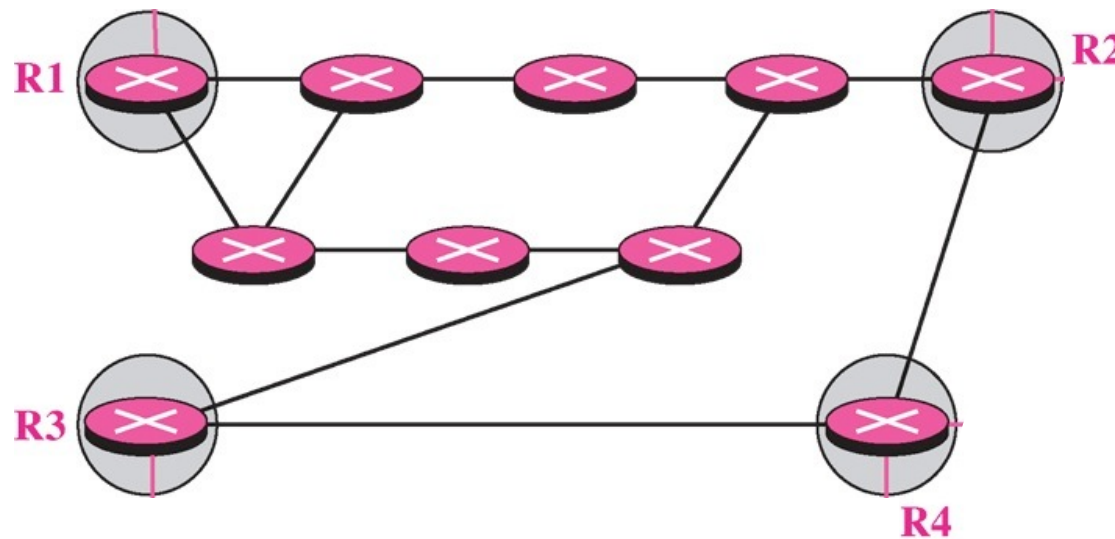
- Once the shared tree is in place:
 1. Host sends packet to group G on its LAN
 2. It is received by the Designated Router (here R1), besides any other local host
 3. The DR tunnels it to the RP, encapsulated in a normal *unicast* IP packet
 4. RP receives the packet, opens it, and forwards it down the shared tree (here to R2, then R4 and R5)



- notice that we can assume that routers between DR and RP are not participating to the tree (they may be not implementing multicast at all, as often happens on the Internet); only DR needs to know which is the RP

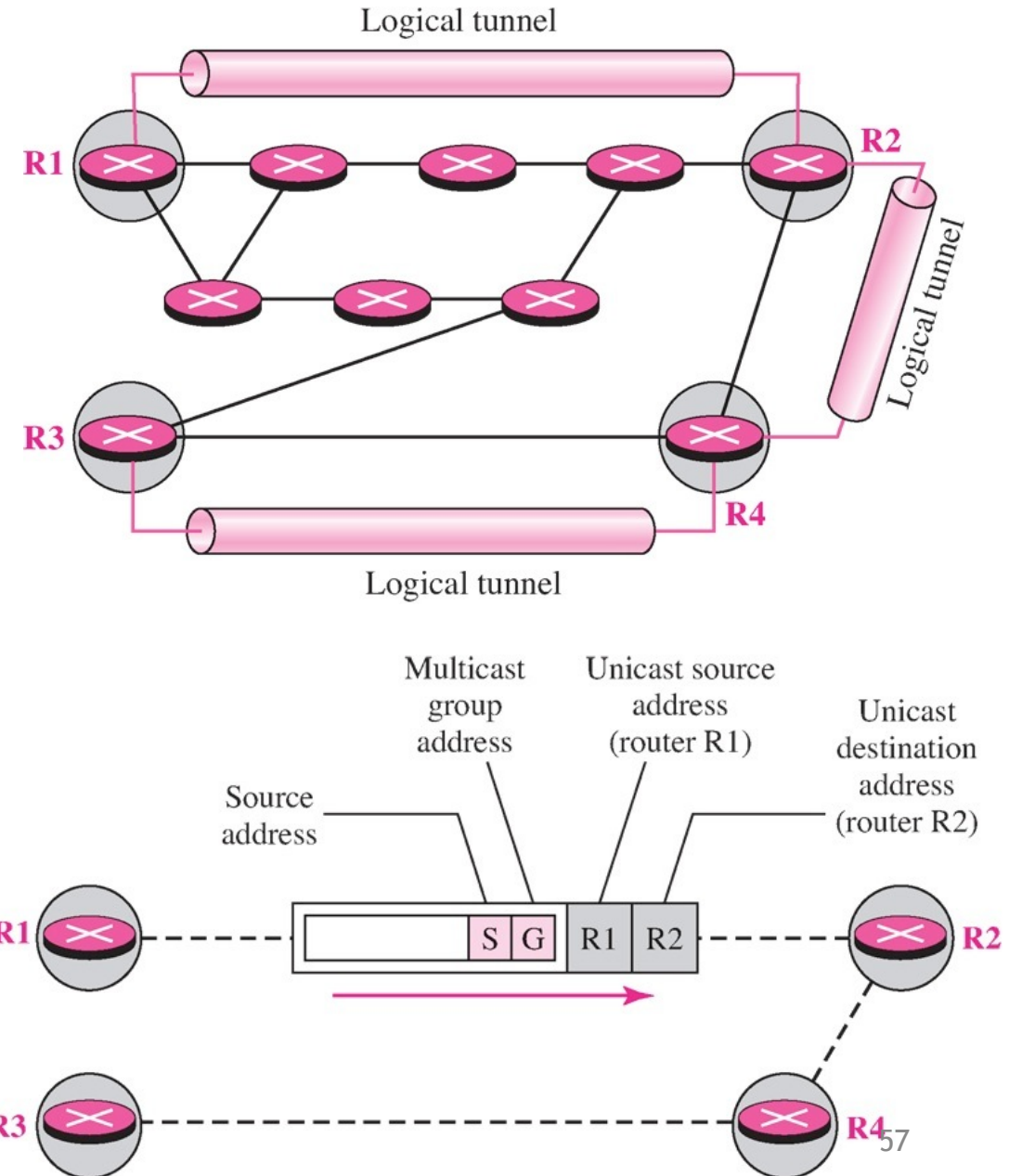
Multicast Backbone (MBONE)

- As per today, only a small fraction of routers implement multicast routing
 - Administrators are not keen to admit multicast traffic due to their cost in routers
- Often these routers are not contiguous, i.e., are connected with regions which are not multicast-aware
 - aside: R1, R2, R3, R4 are multicast-aware routers, but all the others are not (they just drop all multicast traffic)



Multicast Backbone (MBONE)

- “Temporary” solution: create some unicast **tunnel** between the multicast router, crossing through non-multicast router
- Tunnel are logical links, and act as a *backbone* for multicast, called **MBONE = Multicast backbone**
- Multicast packets are encapsulated inside normal unicast IP packet, and can move through the multicast-unaware routers as usual.
- Multicast routing protocols can run over the MBONE



Multicast Backbone (MBONE)

- MBONE (and multicast in general) are not very widespread
 - accounting issues, traffic management, traffic on the providers...
 - More load on routers
 - largest event ever transmitted live: Rolling Stones concert in Dallas, 1994 (50.000 clients)
- MBONE is obsoleted; going to be replaced by IPv6 with PIMv6
- Still, multicast is adopted within single organisations (at levels of LANs and AS), e.g. for carrying IPTV content within an hotel/hospital/provider domain...

Summary

- We have looked at the issues of scalability in routing in the Internet
- We have seen IPv6
- We have discussed Multicasting
- ~~We have discussed Mobile IP~~